ENGR 212 Laboratory Experiment 9

Informal Lab Report

Capacitors and RC Circuits

Objectives:

1. Study circuits with both resistive elements and capacitors. The main focus is on the RC time constant and how it affects capacitor charging and discharging.

Materials:

1. Agilent DSO3202A Oscilloscope
2. Agilent 33120A or 33220A 15MHz Function/Arbitrary Waveform Generator
3. One breadboard
4. Resistors (1 kΩ, 1.2 kΩ, 1.5 kΩ, 1.8 kΩ, and 2 kΩ), capacitors (0.01 μF and 1 μF).

Capacitor charging/discharging in an RC circuit:

The RC circuit and dc excitation source V_s are shown in Fig. 1. From KVL,

$$R\,i(t) + v_C(t) = V_s \qquad (1)$$

From the *v-i* relationship of the capacitor,

$$i = C\frac{dv_C}{dt} \qquad (2)$$

Combining (1) and (2) with some arrangement leads to

$$\frac{dv_C}{(v_C - V_s)} = \frac{-dt}{RC} \qquad (3)$$

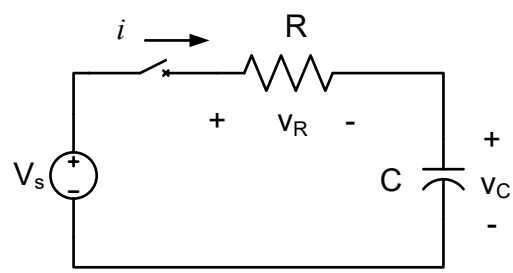

Fig. 1, RC circuit with the excited voltage source V_s

The voltage across the capacitor can be obtained by integrating both sides of (3) followed with some manipulations:

$$v_C(t) = V_s + Ke^{\frac{-t}{RC}}, \qquad (4)$$

where K is a constant. Assuming there is no charge stored in the capacitor before the switch in Fig. 1 is closed, $v_C(0)$ is zero. Substituting t = 0 into (4) leads to

$$v_C(0) = V_s + K = 0. \qquad (5)$$

Therefore, K = $-V_s$. The voltage across the capacitor charged by the dc voltage source can be expressed as

$$v_C(t) = V_s\left(1 - e^{\frac{-t}{RC}}\right). \qquad (6)$$

Substituting $t = RC$ into (6), $v_C(t)=0.632*V_s$. Here the product of RC is called the time constant of the circuit.

To derive $v_C(t)$ for discharging the capacitor, one can set $V_s = 0$ in Fig. 1. As a result, the solution of $v_C(t)$ for discharging can be obtained from (4) by setting $V_s = 0$ as

$$v_C(t) = Ke^{\frac{-t}{RC}}, \qquad (7)$$

where K is a constant. Assuming that capacitor discharging starts at t = 0 with the initial condition $v_C(0) = V_0$, The constant K in (7) is equal to V_0. Therefore, $v_C(t)$ for capacitor discharging can be written as

$$v_C(t) = V_0 \cdot e^{\frac{-t}{RC}}. \tag{8}$$

When t = RC, $v_C(t) = 0.368 * V_0$.

Experimentation:

1. Build the RC circuit in Fig. 1 on the breadboard with R = 1 kΩ and C = 0.01 μF.
2. Set up the function generator to "High Impedance" mode as described in the following:
    a. For **33120A** function generator:
        i. Push "Shift" then "Menu".
        ii. Push "<" or ">" button to get "D: SYS MENU".
        iii. Push "∧" or "∨" button to get "50 OHM".
        iv. Push "<" or ">" button to get "HIGH Z".
        v. Push "Enter".
    b. For **33220A** function generator:
        i. Push "Utility" and select "Output Setup".
        ii. Select "High Z" and then Push "Done".
3. Setup the function generator to produce a 5 kHz square wave with 1 V peak-to-peak, 0.5 V dc offset, and 50% duty cycle as the excitation voltage source V_s.
4. Connect V_s to CH1 of the oscilloscope to verify the signal. Adjust the Time/div on the oscilloscope to display slightly more than 1 period of the signal. Set up the V/div on the oscilloscope to 500 mV/div.
5. Connect v_C in Fig. 1 to CH2 of the oscilloscope.
6. Connect V_s to the circuit board. You should see a periodical charging/discharging curve on CH2.
7. Push the STOP button to hold the signals.
8. Find the cursor menu on the oscilloscope. Use the **tracking cross-hair cursors** (see end of lab if you are using the Agilent DSO3202A oscilloscope) and adjust the position of the cursor A cross-hairs to the minimum (about 0 v) of v_C on CH2 so that they are appoximately at the point where v_C is about to rise. Position the cursor B cross-hairs at the point where $v_C \approx 0.632 * V_s|_{max}$ and within the same period as the cursor A cross-hairs. Record Δt displayed on the oscilloscope in Table 1 as the charging time constant.
9. Adjust the position of the cursor A cross-hairs to the maximum (about 1 v) of v_C on CH2, close to the point where v_C is about to fall. Adjust the cursor B cross-hairs to the point where $v_C \approx 0.368 * V_s|_{max}$ and within the same period as the cursor A cross-hairs but after the cursor A. Record Δt displayed on the oscilloscope in Table 1 as the discharging time constant.
10. In case you lose the CH2 signal while you are adjusting the cursors, push the RUN button to re-capture the signals and then press STOP to continue step 8 and 9.

11. Disconnect V_s from the circuit board and change the resistor R to 1.2 kΩ, 1.5 kΩ, 1.8 kΩ, and 2 kΩ. Repeat step 6 to 10 with different values of resistance.
12. Generate a plot of time constants versus resistances from your data. Use this plot to determine the capacitance.
13. Disconnect V_s from the circuit board and change the capacitor to 1 μF to **repeat the experiment**. Be sure to connect the polarity of the capacitor correctly. Now you are using a bigger capacitor, and thus a longer time constant. You must change the frequency of the square wave so the capacitor can be charged and discharged "long enough". In general, seven time constants are relatively "long enough" for fully charging and discharging. Therefore, you would need to select a square wave with overall period greater than 14 time constants.

Table 1, Data Sheet for capacitor charging/discharging experiment

V_{PP} = _____; Offset = _____; Duty Cycle (%) = _____; Frequency (Hz) = _____		
C = _____		
R	*Charging Time Constant*	*Discharging Time Constant*
1 kΩ		
1.2 kΩ		
1.5 kΩ		
1.8 kΩ		
2 kΩ		

To use tracking cross-hair cursors (from page 99 of Oscilloscope manual)

You can set up one or two manually adjustable, tracking cross-hair cursors to make amplitude (vertical) and time (horizontal) measurements at different points of a selected channel's waveform.

1. Press **Cursors.**
2. In the Cursors menu, press **Mode** until "Track" is selected.
3. Press **Cursor A** to select the channel on which to make the measurement (or "None" to turn off the cursor).
4. Press **Cursor B** to select the channel on which to make the measurement (or "None" to turn off the cursor).
5. To adjust the cursors:
    a. Select **CurA** and turn the **Entry** knob to adjust the "A" cursor.
    b. Select **CurB** and turn the **Entry** knob to adjust the "B" cursor.

The A cursor values displayed are:

- A->X.
- A->Y.

The B cursor values displayed are:

- B->X.
- B->Y.

If both A and B cursors are used, these values are also displayed:

- DeltaX – difference between CurA and CurB time values.
- 1/DeltaX – shows the frequency associated with the time value difference
- DeltaY – difference between CurA and CurB amplitude values.